Substituting in (1) we have

$$r_{XY}^2 = \frac{0.36 \times 25}{16}$$

$$r_{XY} = \sqrt{\frac{0.36 \times 25}{16}} = 0.75$$

Example 5.9

If two regression coefficients are $b_{YX} = \frac{5}{6}$ and $b_{XY} = \frac{9}{20}$, what would be the value of r_{XY}?

Solution:

The correlation coefficient $r_{XY} = \pm\sqrt{(b_{YX})(b_{XY})}$

$$= \pm\sqrt{\frac{5}{6} \times \frac{9}{20}} = \sqrt{0.375} = 0.6124$$

Since both the signs in b_{YX} and b_{XY} are positive, correlation coefficient between X and Y is positive.

Example 5.10

Given that $b_{YX} = -\frac{8}{7}$ and $b_{XY} = -\frac{5}{6}$. Find r?

NOTE

The sign of the corelation coefficient will be the signs of the regression coefficients.

Solution:

$$r_{XY} = \pm\sqrt{(b_{YX})(b_{XY})}$$

$$= \sqrt{-\frac{8}{7} \times -\frac{5}{6}} = \sqrt{\frac{20}{21}} = -0.9759.$$

Since both the signs in b_{YX} and b_{XY} are negative, correlation coefficient between X and Y is negative.

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